

THE INCA TRAIL

THE INCA TRAIL | PURPLE GOLD

SOURCING PARTNERSHIP:	Native Shuar Indigenous Farming Communities (Napurak, Los Bayanes)
GENETIC ORIGIN & TERROIR:	Zamora-Chinchipec, Ecuador - The Birthplace of Cacao (1,000 - 1,200m a.s.l.)
PRIMARY APPLICATION:	Premium Functional Food & Nutraceutical Grade Ingredients
COLLABORATION TARGET:	Standardized Active Functional Ingredients for Global R&D; & Brands

1. ORIGIN, GENETICS & POST-HARVEST LOGISTICS

To capture and freeze the native biochemistry of the living bean, harvesting is synchronized with processing. Pods are opened strictly in the morning. By noon, the wet pulp ('baba de cacao') is collected in refrigerated, insulated containers and transported to our processing facility within a maximum of 3 hours. The entire post-harvest cycle - from pod breaking to final vacuum sealing - is completed in less than 7 hours. This rapid, climate-controlled handling completely prevents any spontaneous or uncontrolled fermentation, preserving the original cellular integrity and high-value bioactive compound concentrations of the fresh seed.

As a dedicated B2B raw material and innovation partner, our primary mission is to supply functional food and nutraceutical brands with this pristine, biochemically intact raw ingredient. By operating this meticulous direct-sourcing model, we provide the ultimate bioactive building blocks for advanced functional collections, moving beyond conventional trade to a true co-creation partnership.

2. FAST MODIFIED ATMOSPHERE DEHYDRATION (FMAD) TECHNOLOGY

Traditional sun drying and open-air fermentation trigger intense enzymatic and oxidative reactions. These processes degrade up to 70% of the active polyphenolic complex and evaporate volatile, thermosensitive neurotransmitters such as Phenylethylamine (PEA). To permanently lock in the 'living matrix' of the fresh bean, we utilize proprietary Fast Modified Atmosphere Dehydration (FMAD) technology. Our engineered drying protocol applies a rapid, low-temperature electromagnetic thermo-shock under controlled vacuum.

This precise thermal and pressure impulse instantly denatures the polyphenoloxidase (PPO) and peroxidase (POD) enzymes, arresting enzymatic browning and halting the oxidation of monomeric flavanols. Consequently, the vibrant, royal purple color of the anthocyanins is entirely preserved. In the final stage, an adiabatic flash-cooling is executed. This causes residual intracellular moisture to boil instantly at low temperatures. This explosive steam release gently expands the cotyledon cell structure, providing a unique, light crispiness ('Glassy Crunch') and achieving an ultra-low, shelf-stable moisture content of 6-7% without any heat damage.

3. PREMIUM RAW MATERIALS & INGREDIENTS SPECIFICATIONS

Below is the primary premium raw material produced utilizing our technology, now delivered to global partners as an active functional ingredient.

PRODUCT 1: Whole Purple Cacao Beans in Shell (Ready-to-Eat)

Un-fermented, non-roasted whole cacao beans dried via FMAD. The intact shell (husk/veleta) acts as a sterile protective envelope, preserving the internal cotyledons from oxygen. The shell is fully retained, serving as a rich source of prebiotic dietary fiber and highly bioavailable trace minerals. It possesses a distinct glassy-crunch texture.

Parameter	Specification	Functional & R&D; Value
Physical Form	Whole beans with intact shell	Natural protective barrier preserving inner fats and delicate compounds from oxidation
Moisture Content	< 7.0% (dry basis)	Ultra-low moisture with a unique, light 'Glassy Crunch' texture
Fat Content (with shell)	35.0% - 37.0% (natural fat)	Rich in native lipids acting as a natural carrier for fat-soluble compounds
Theobromine / Caffeine (T/C)	5.7 - 6.5 (Genetically Pure)	Pure Cacao Nacional genetic signature; delivers smooth, jitter-free ceremonial focus
Sensory Profile	Intense raw cocoa, pleasant astringency, fruity undertones	Bold, native floral and fruit notes characteristic of Zamora-Chinchipe terroir
Microbiology	Commercially sterile, yeast/mold negative	FMAD process ensures immediate sterility, making the bean safe to consume whole

Sent Samples & Formulation Ideas (Demonstrating Versatility)

Along with our primary raw material (Whole Purple Cacao Beans), we have sent several conceptual samples to demonstrate the diverse application possibilities of this unique ingredient. These are intended purely as inspiration for R&D; teams to explore how our FMAD raw material can be formulated:

- **100% Pure Purple Cacao Paste (Mass):** Cold stone-milled strictly below 42°C with the shell fully retained and processed together with the cotyledons, demonstrating how the raw bean can be converted into a rich, bio-active mass without any alkalization or thermal damage.
- **Inca (Synergistic Functional Bar):** A cold-processed chocolate bar concept combining our 100% pure purple paste with panela, cocoa butter, and optimized botanical synergies (including dried naranjilla, black pepper, and warm spices) to show sensory and bioavailability integration.
- **The Bio-Active Crunchy Snack:** Whole FMAD Purple Cacao Beans in Shell coated with a synergistic botanical glaze to illustrate how the whole bean can be presented as a finished functional food.

4. BIOCHEMICAL COMPOSITION & COMPARATIVE PROFILE

With our direct Shuar-harvested Cacao Nacional (Fino de Aroma) and INIAP laboratory baselines for fresh Nacional raw material (Napurak 25-0803), the biochemical profile of our dry product (moisture content 6-8%, output ratio 1:3, and preservation $\geq 90\%$) is specified below:

1. Alkaloid Ratio (Nacional Genetic Marker)

In contrast to hybrids (CCN-51, Forastero), pure Nacional is characterized by a moderate theobromine content and low caffeine (T/C ratio $\approx 5.7 - 6.8$):

- **Theobromine:** 2.15% - 2.40% (or 21.5 - 24.0 mg/g / 2150 - 2400 mg/100g). In fresh Nacional raw material, 2.40% theobromine on dry matter was recorded. During our gentle FMAD drying, without the prolonged water washing of fermentation, this value is preserved virtually without loss.
- **Caffeine:** 0.35% - 0.40% (3.5 - 4.0 mg/g / 350 - 400 mg/100g), compared to 0.42% in the fresh bean.

2. Polyphenolic & Flavanol Complex

Due to the absence of fermentation and the instant denaturation of the PPO enzyme by thermo-shock under vacuum, monomers do not oxidize into insoluble brown tannins:

- **Total Polyphenols:** 80 - 95 mg GAE/g (or 8.0 - 9.5 g / 100g). In fresh Nacional (Napurak), it is 89.64 mg GAE/g. With a 90% preservation rate, our dry bean holds ≈ 80.7 mg GAE/g, which is 2.5 to 3 times higher than traditional fermented cacao (30 - 35 mg GAE/g).
- **(-)-Epicatechin (Key Monomer DP1):** 17.0 - 19.5 mg/g (or 1700 - 1950 mg/100g). In the fresh Nacional bean, the epicatechin level is 19.02 mg/g.
- **Total Flavanols (Oligomers DP1-DP7):** 55 - 70 mg/g (5.5 - 7.0 g / 100g).
- **Procyanidins / Oligomers (DP2-DP7):** 38 - 52 mg/g.

4. BIOCHEMICAL COMPOSITION & COMPARATIVE PROFILE
(CONTINUED)

Bioactive Marker	Traditional Fermented Nacional (Export)	Purple FMAD Nacional (90% Retention)
Color / Cut Test	Brown / Tan (oxidized, condensed tannins)	Deep Purple / Violet (pristine anthocyanins DP1-DP7)
Theobromine	1.4% - 1.7%	2.15% - 2.40%
(-)-Epicatechin (DP1)	1.0 - 4.8 mg/g	17.0 - 19.5 mg/g
Total Polyphenols (Folin-C.)	30 - 35 mg GAE/g	80 - 95 mg GAE/g
Total Flavanols (DP1-DP7)	10 - 20 mg/g	55 - 70 mg/g

Our non-fermented, non-roasted FMAD Purple Nacional acts as a natural food-state nutraceutical, delivering extremely high levels of unaltered bioactive compounds. Instead of consuming synthetic supplements or extracted pills, consumers can obtain therapeutic concentrations of active monomers and oligomers directly from the raw whole bean. For vascular health, the targeted dose of pure monomeric epicatechin is approximately 200 - 500 mg. Utilizing our FMAD Purple Cacao, this target is easily achieved with just 10 - 15 g of whole beans or powder, compared to 50 - 100 g of standard industrial fermented cocoa powder. This provides a clean-label, whole-food alternative to pill delivery systems while completely preserving the natural matrix of the seed.

5. METABOLIC SYNERGY & SYSTEMIC BIOAVAILABILITY

Cacao is one of the most chemically complex and powerful whole-food superfoods on the planet, acting as an organic 'polypill' without requiring synthetic extraction. To maximize the therapeutic impact of these complex flavanols, our raw ingredients can be paired with synergistic botanical elements that optimize physiological absorption:

- **Piperine Bio-Enhancement:** Active flavanols (epicatechin) undergo rapid glucuronidation in the digestive tract. Micro-doses of piperine temporarily inhibit these intestinal and liver enzymes, dramatically boosting epicatechin absorption and preventing the rapid breakdown of Phenylethylamine (PEA).
- **Organic Acid Decoupling:** Natural citric and ascorbic acids lower the gastrointestinal pH, preventing minerals like magnesium, iron, and zinc from binding with phytic acid, rendering them fully bioavailable.
- **Active Sodium-Glucose Transport (SGLT1):** Minimal carbohydrate-mineral additions (from panela and sea salt) activate the SGLT1 pump in the gut wall, actively pulling water-soluble flavanols directly into the bloodstream.
- **L-Theanine Modulation:** Combining guayusa with cacao introduces L-theanine. This crosses the blood-brain barrier, stimulating alpha-wave activity to produce a deep, calm focus and smoothing out the vascular effects of cocoa's methylxanthines.

6. SOMATOSENSORY TASTING PROTOCOL & COGNITIVE CHALLENGE

We invite R&D; and product development teams to run these two somatic experiments to personally experience the unmatched biochemical and sensory potency of this raw material. If you do not have physical samples on hand, please request our Somatic Sample Kit at the address listed below.

EXPERIMENT 1: The 5-Bean Somatic Challenge (10g Whole Bean Serving)

Instructions: Consume exactly 5 whole FMAD Purple Cacao beans (approximately 10 grams total) without peeling. Eat them together with their crunchy and sweetish husk (veleta/shell) which acts as a prebiotic carrier. Chew slowly and thoroughly to release the trapped volatile matrix.

Somatic Experience: Active compounds are absorbed, starting with theobromine which acts as a potent smooth-muscle bronchodilator, making breathing immediately feel deeper and highly oxygenated. Simultaneously, un-roasted Phenylethylamine (PEA) crosses the blood-brain barrier, stimulating dopamine release and producing a noticeable rise in mental clarity, emotional empathy, and a warm sense of well-being. Finally, active monomeric epicatechin stimulates endothelial nitric oxide synthase, leading to vascular dilation and a distinct warmth spreading through the body.

EXPERIMENT 2: The Synergistic Bar Degustation (20g - 30g Functional Sample 'Inca')

Instructions: Slow-melt a 20g - 30g portion of our functional chocolate bar concept marked 'Inca' (combining our FMAD Purple Paste and optimized botanical synergies) on your tongue. Do not chew; let the lipids coat your mouth to slowly release the functional matrix.

Somatic Experience: In this experience, the sodium-glucose SGLT1 transport pump and piperine bio-enhancer work in tandem, amplifying the absorption of active compounds. This leads to an intense, heart-opening cardiovascular warmth and a deep, jitter-free 'calm focus' state that sustains comfortably for 4 to 6 hours, replicating the exact, highly active spiritual and cognitive states that were sacredly preserved and guarded by our ancestors.

7. OUR CO-CREATION MANIFESTO: PRESERVING THE PRIMORDIAL CODE

Zamora-Chinchipe, Ecuador is scientifically recognized as the genetic birthplace of all world cacao, and our endemic Cacao Nacional represents a direct, non-hybridized descendant of the very first domesticated trees on Earth. This precious primordial chemistry is carefully preserved by traditional indigenous Shuar farming communities through their ancestral, highly biodiverse forest planting system known as 'chakra'.

By merging this sacred genetic heritage with our proprietary Fast Modified Atmosphere Dehydration (FMAD) technology, we physically freeze and preserve the original, unaltered 'source code' of the living cacao seed. Our meticulous, climate-controlled 'tree-to-bean' logistics completes the entire harvest-to-vacuum-sealing process in less than 7 hours, preventing any spontaneous fermentation or oxidative degradation.

This entire process is fully transparent and traceable: every single bean can be linked back to a specific chakra, an individual tree, and the Shuar family who nurtured it. This direct-trade model guarantees an exceptionally high added value for every participant in the value chain - from the native farming communities receiving premium-above-market compensation, to the B2B manufacturer, the brand owner, and ultimately, the conscious consumer who experiences the unmatched physical and cognitive power of living, active cacao.

8. COMMERCIAL OPERATIONS & B2B PARTNERSHIP

We operate strictly under forward contracts and customized specifications to provide high-value, active cacao ingredients to innovative brands globally.

Operational Parameter	Specification	Terms & Customization Details
Minimum Order Quantity (MOQ)	60 kg (Product 1: Whole Beans)	Optimized for pilot runs and formulation R&D; trials.
Fulfillment Lead Time	15 work days	Subject to order volume, seasonal factors, and current stock availability.
Sourcing Limit (2027 Forecast)	Up to 2,400 kg wet pulp / month (yielding ~800 kg dry FMAD / month)	All large volumes are secured via annual forward contracts with pre-allocated Shuar chakra lots.
Shelf-Life & Packaging	24 Months	Sealed under deep vacuum. Store in dry, cool conditions away from direct sunlight.
TDS & Lab Customization	On-Demand HPLC Profiling	We are owners of the FMAD technology. We can customize drying profiles and run certified HPLC tests (for flavanols, PEA, heavy metals) directly tailored to the Technical Data Sheet (TDS) of your R&D; department.
Shipping Terms	EXW Zamora / FOB Guayaquil	Air freight recommended for R&D; batches. Sea freight available in refrigerated containers.

REQUEST A SOMATIC SAMPLE KIT

Experience the power of living biochemistry. To request your standardized B2B sample package (including whole FMAD Purple Cacao beans and trial paste samples), contact our innovation desk at:
Email: info@incatrailfoods.com | Website: incatrailfoods.com

CRITICAL R&D; PROCESSING WARNING: To protect the fragile polyphenolic chains and volatile PEA in these ingredients, traditional high-temperature open-air conching, roasting, or chemical alkalization must be completely avoided. Processing must remain closed-loop and cold-formulated (strictly below 42°C) to maintain active functional integrity.